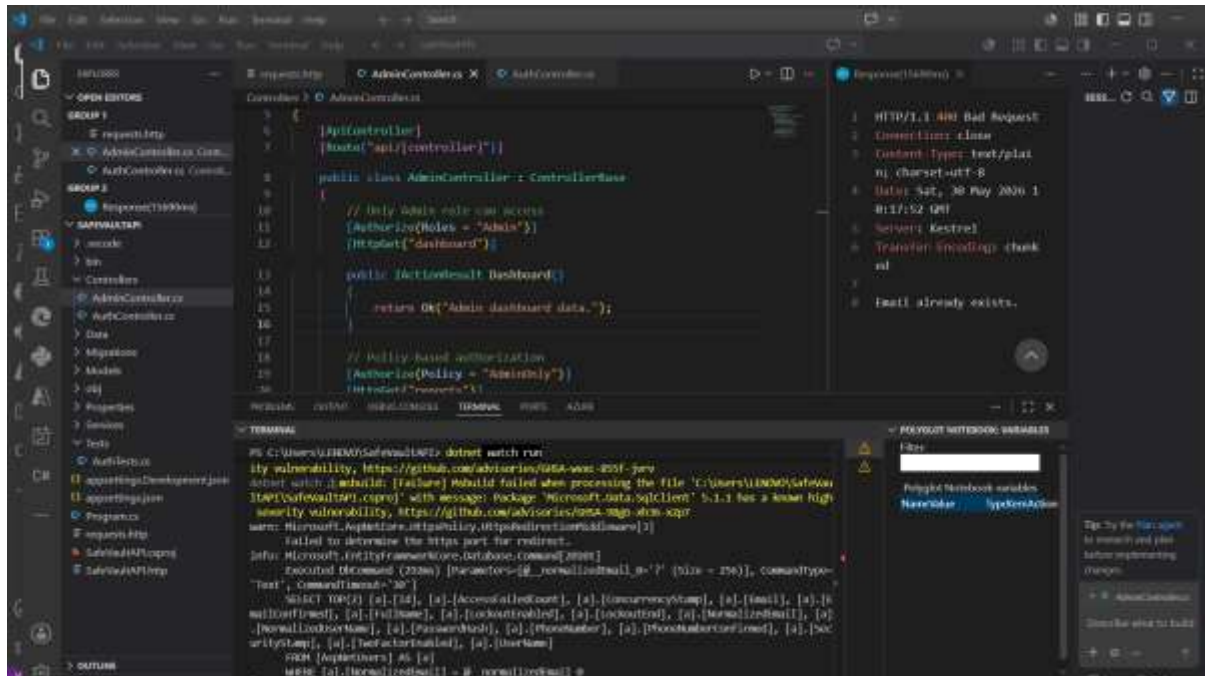


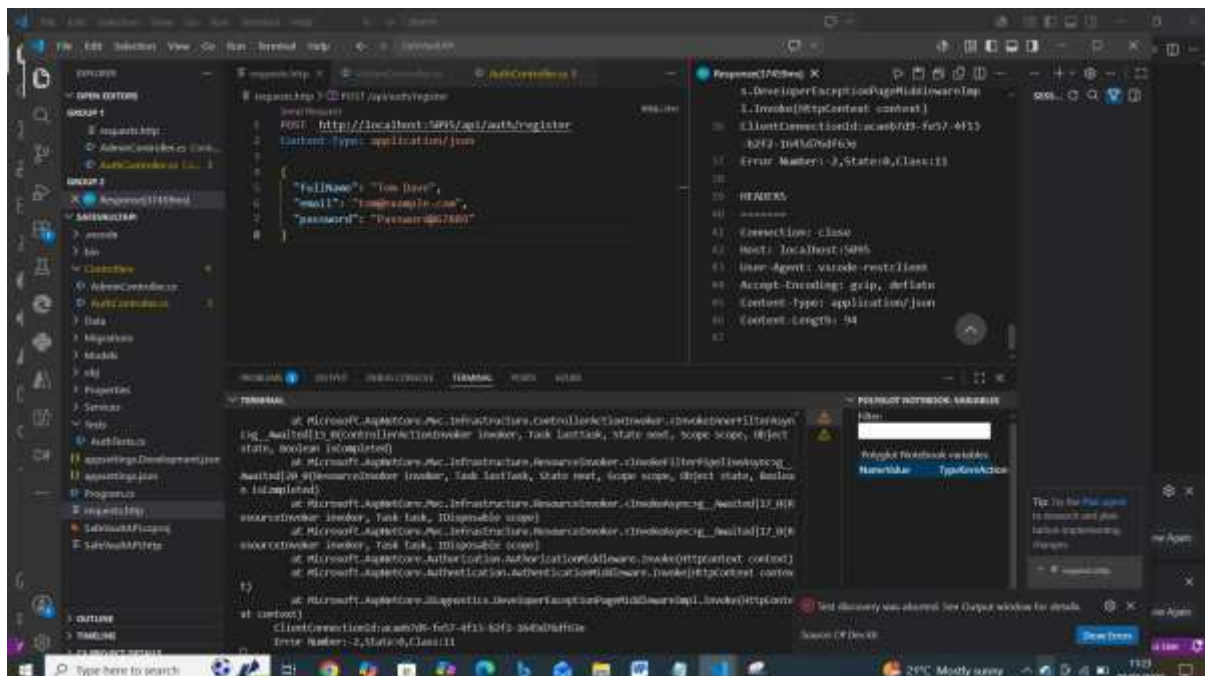
Kayode Oladipo –Debugging assignment

TESTING AND DEBUGGING

Rest client was installed from the visual studio code IDE extension. The application was tested using watch run.

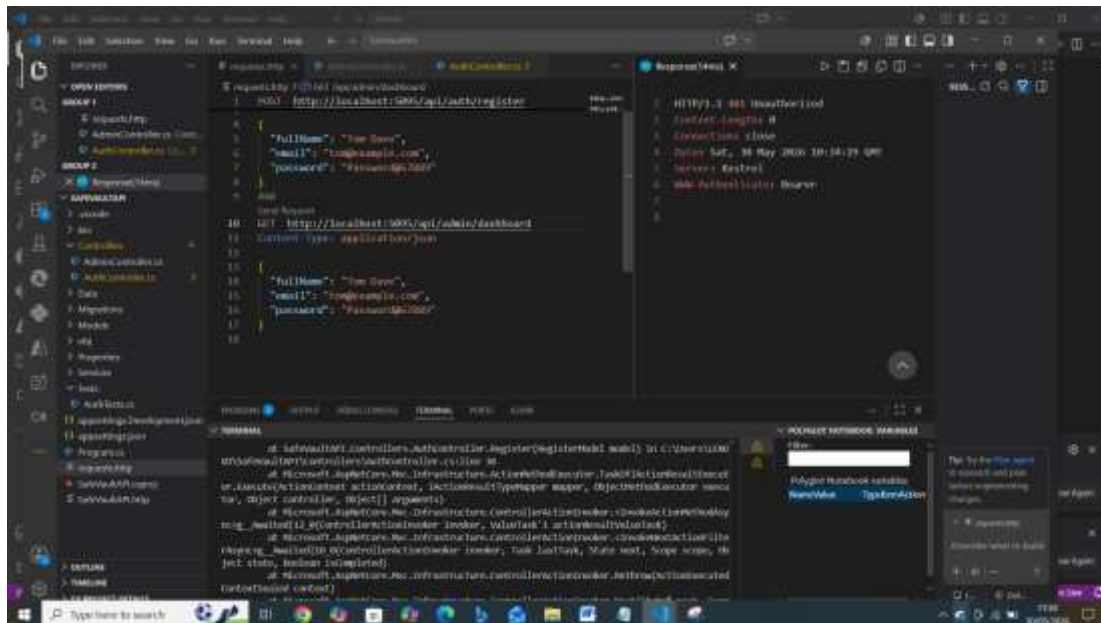


The application runs successfully.



Initially after attempting to register a user the headers were displayed because I have not login.

Kayode Oladipo –Debugging assignment



I made this request in my application:

GET <http://localhost:5095/api/admin/dashboard>

Content-Type: application/json

```
{
  "fullName": "Tom Dave", "email": "tom@example.com", "password":
  "Password@67889"
}
```

This is what I am getting

HTTP/1.1 401 Unauthorized Content-Length: 0 Connection: close Date:
Sat, 30 May 2026 10:34:29 GMT Server: Kestrel WWW-Authenticate:
Bearer.

This demonstrates that the data I stored in the register is protected and needs authorisation may be a bearer 's token to allow me to access the data.

The response:

HTTP/1.1 401 Unauthorized
WWW-Authenticate: Bearer

my endpoint is protected by JWT Bearer authentication, and my request does not contain a valid access token. The dashboard endpoint that is a JWT token. I know this is the case, so I must

Kayode Oladipo –Debugging assignment

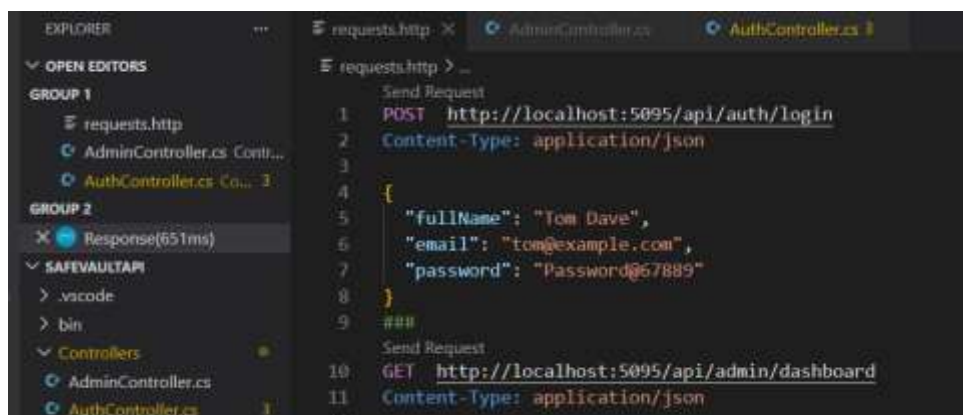
first log in and obtain a token. Then call the dashboard endpoint with:

```
GET http://localhost:5095/api/admin/dashboard
```

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIs...
```

The error is not related to authorization itself. Your JWT token is failing to be generated, because your secret key is too short.

The important part is:



Debugging using AI

Another error occurs as shown below:

```
IDX10720: Unable to create KeyedHashAlgorithm for algorithm 'HS256',  
the key size must be greater than: '256' bits, key has '208' bits.
```

What this means that 'HS256' requires a secret key of at least 256 bits (32 bytes).

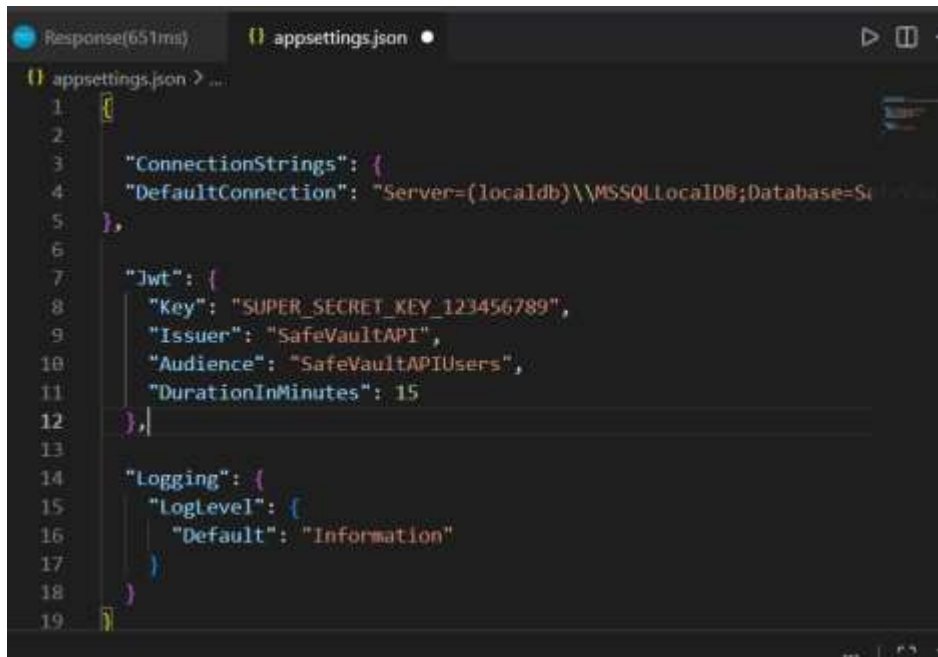
My key is only 208 bits (26 bytes).

Correction

My appsettings.json contains the followings:

```
{  
  
  "Jwt": {  
  
    "Key": "SUPER_SECRET_KEY_123456789",
```

```
}  
  
}
```

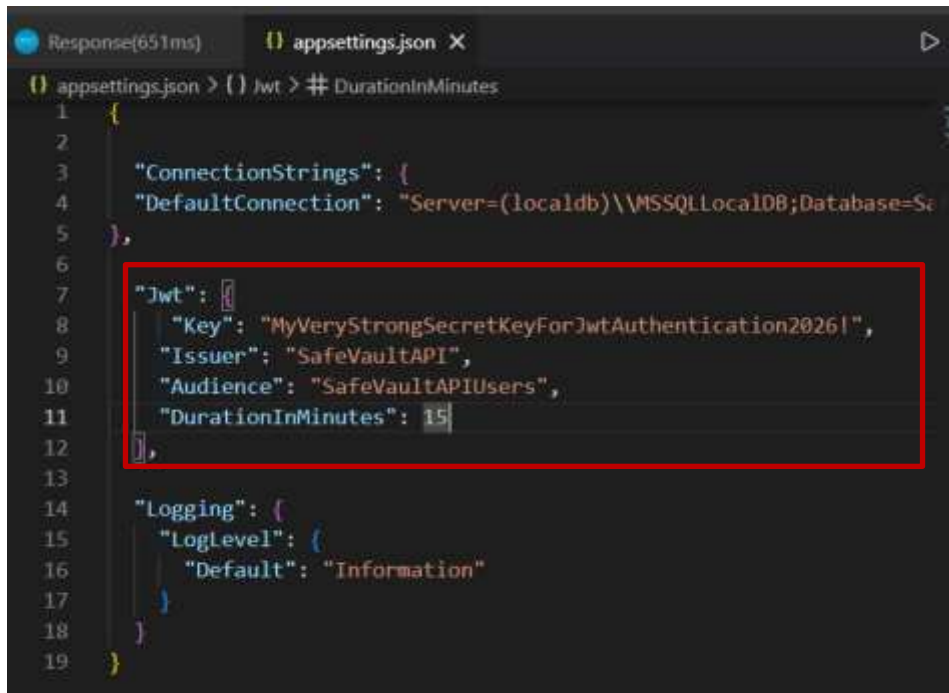


I could also have used the followings in my appsettings.json

```
{  
  
  "Jwt": {  
  
    "Key": "7F4D9A8C2B1E6F3A5D8C9E1B4F7A2D6C9E5F1A8B3D7C4E6F"  
  
  }  
  
}
```

So I corrected it as shown below:

Kayode Oladipo –Debugging assignment

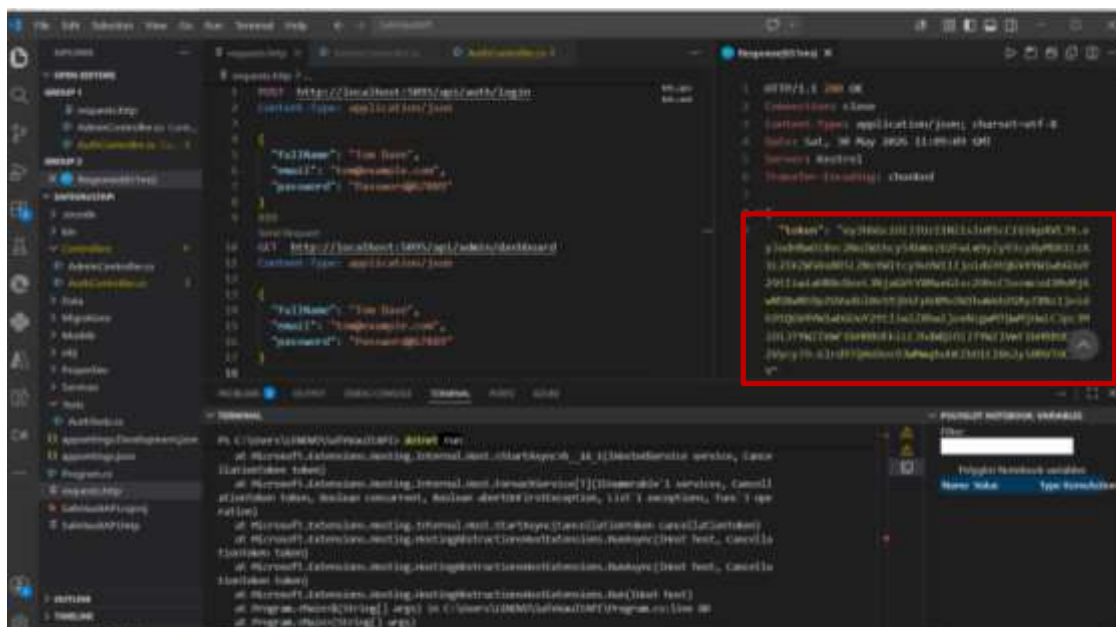


```
Response(651ms) appsettings.json X
appsettings.json > {} Jwt > # DurationInMinutes
1 {
2
3   "ConnectionStrings": {
4     "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=Se
5   },
6
7   "Jwt": {
8     "Key": "MyVeryStrongSecretKeyForJwtAuthentication2026!",
9     "Issuer": "SafeVaultAPI",
10    "Audience": "SafeVaultAPIUsers",
11    "DurationInMinutes": 15
12  },
13
14  "Logging": {
15    "LogLevel": {
16      "Default": "Information"
17    }
18  }
19 }
```

My secret key now contains at least 256 bits (32 bytes).

"Key": "MyVeryStrongSecretKeyForJwtAuthentication2026!".

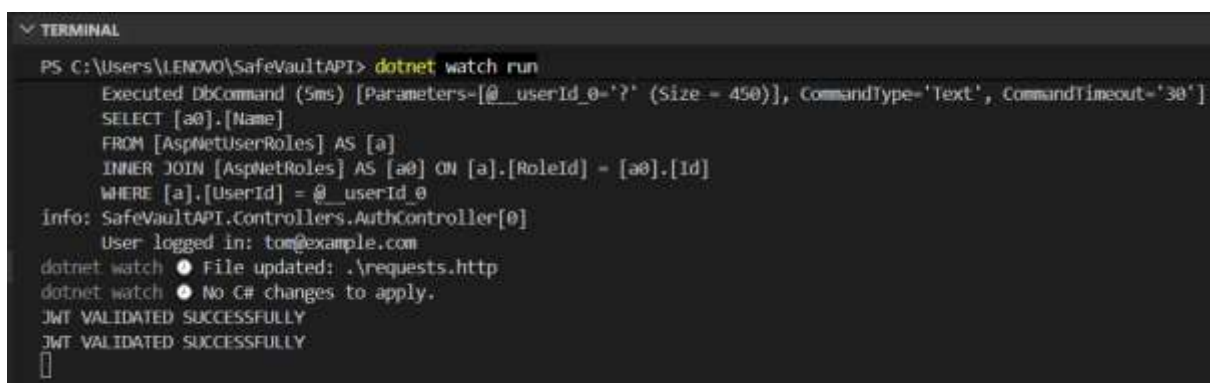
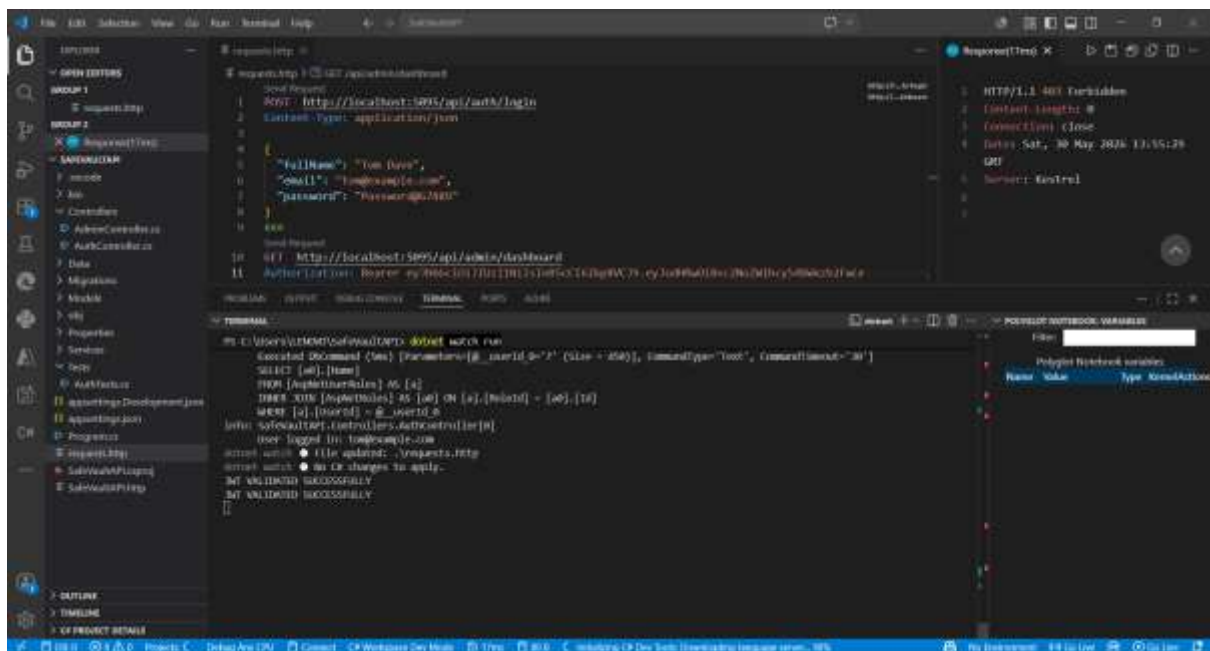
I then made a request to login again and the token was successfully generated.



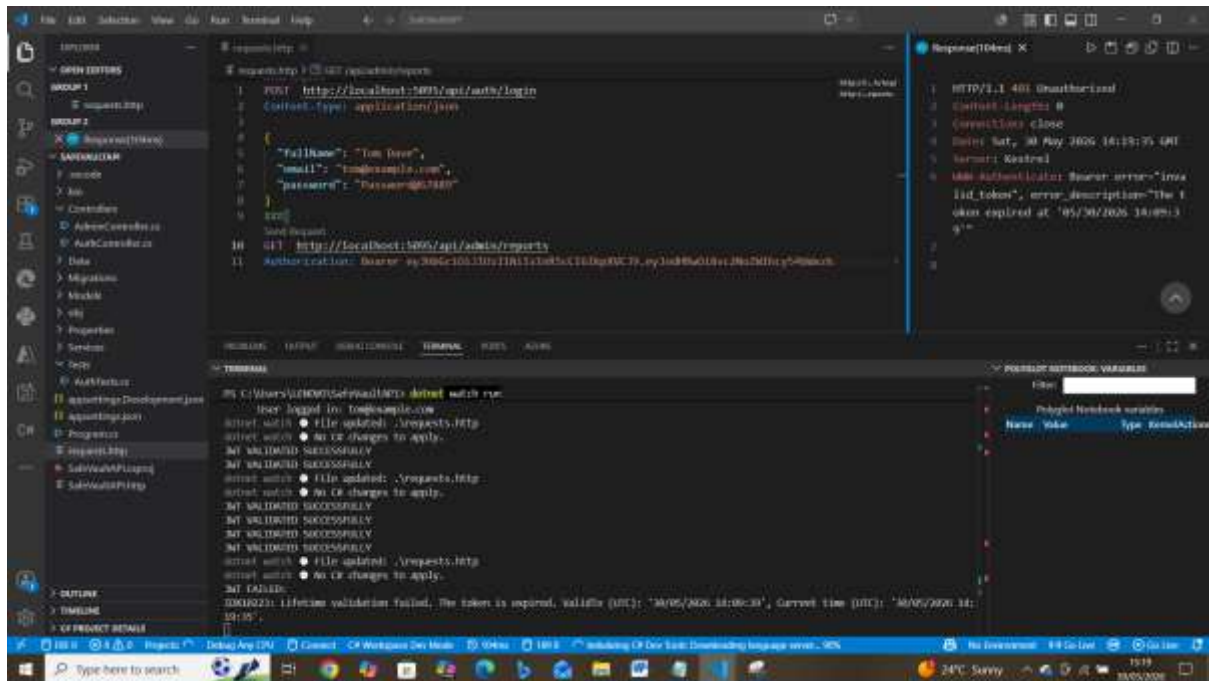
```
File Edit Solution View Run Run and Debug Help
REST Client
1 POST http://localhost:3000/api/auth/login
2 Content-Type: application/json
3 {
4   "username": "John Doe",
5   "email": "john@safevault.com",
6   "password": "Password123!"
7 }
8
9 GET http://localhost:3000/api/auth/dashboard
10 Content-Type: application/json
11 {
12   "username": "John Doe",
13   "email": "john@safevault.com",
14   "password": "Password123!"
15 }
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Kayode Oladipo –Debugging assignment

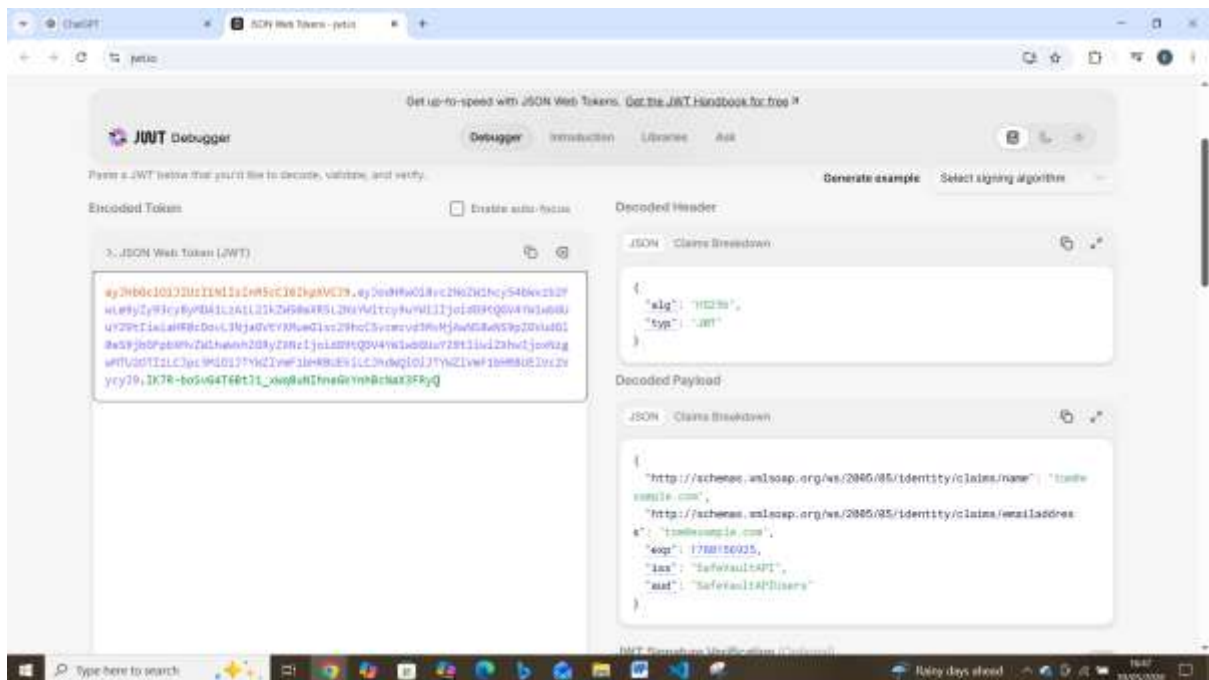
Accessing the dashboard.



Kayode Oladipo –Debugging assignment

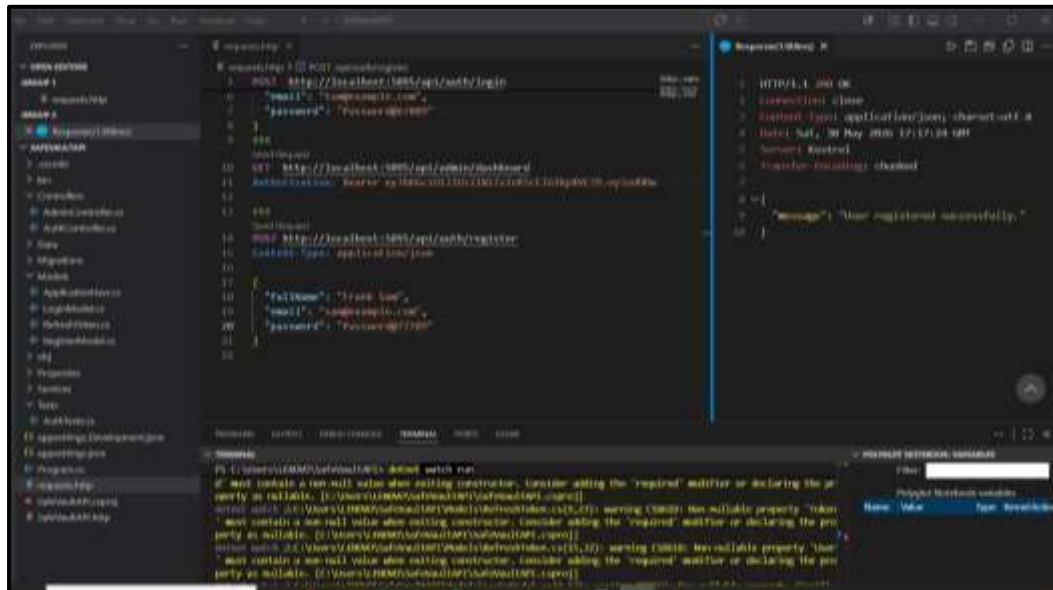


Checking my token to see the payload contains the right data.

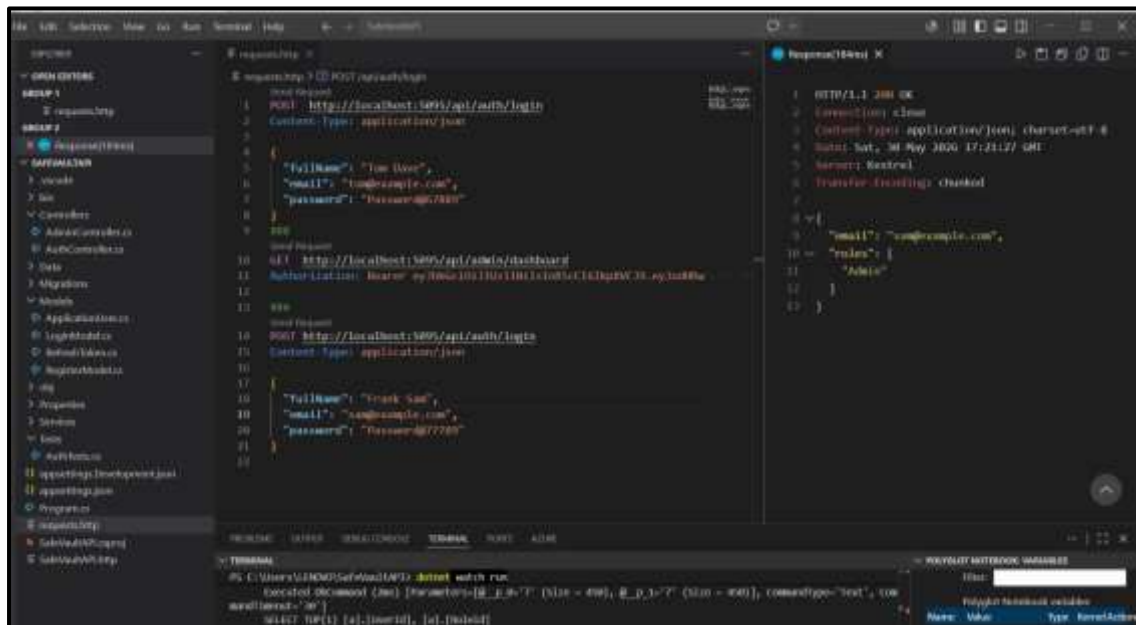


Kayode Oladipo –Debugging assignment

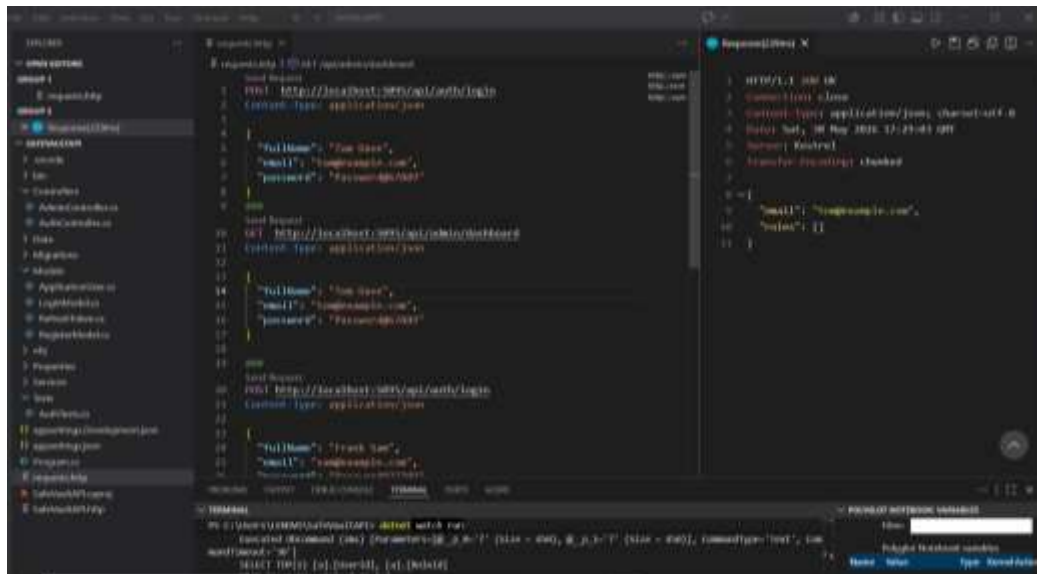
Registering a user.



Login in to make a request.

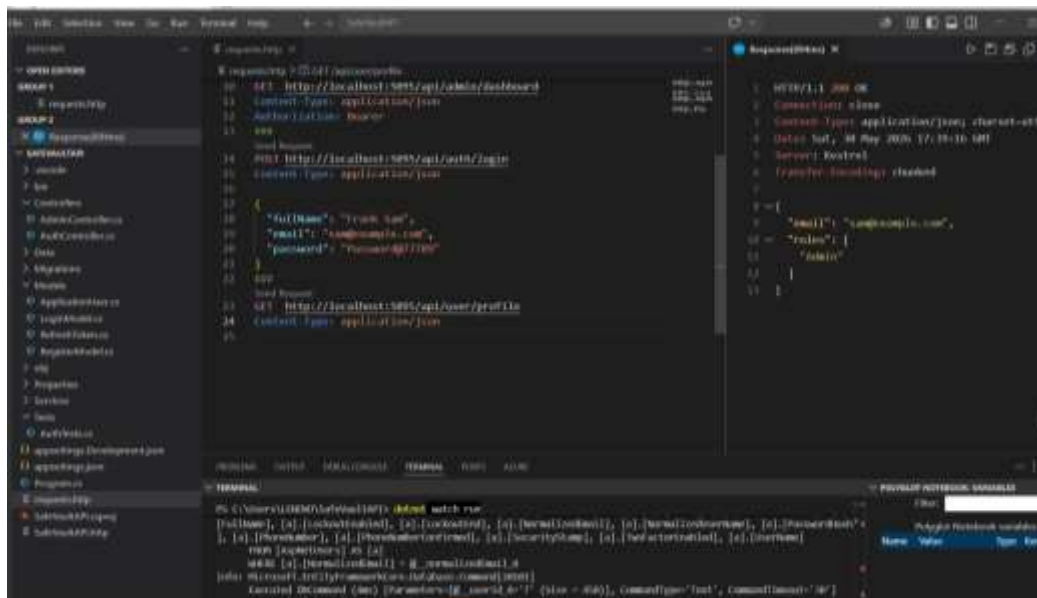


Kayode Oladipo –Debugging assignment



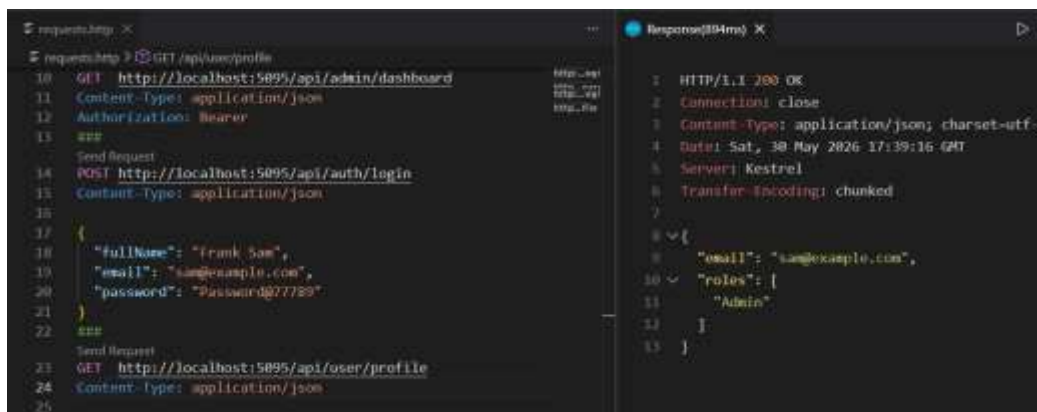
```
1 POST http://localhost:5095/api/auth/login
2 Content-Type: application/json
3
4 {
5   "fullName": "Frank Sam",
6   "email": "sam@example.com",
7   "password": "Password@12345"
8 }
9
10 Send Request
11
12 GET http://localhost:5095/api/admin/dashboard
13 Content-Type: application/json
14
15 {
16   "fullName": "Frank Sam",
17   "email": "sam@example.com",
18   "password": "Password@12345"
19 }
20
21 Send Request
22
23 POST http://localhost:5095/api/auth/login
24 Content-Type: application/json
25
26 {
27   "fullName": "Frank Sam",
28   "email": "sam@example.com",
29   "password": "Password@12345"
30 }
31
32 Send Request
33
34 GET http://localhost:5095/api/user/profile
35 Content-Type: application/json
36
37 {
38   "fullName": "Frank Sam",
39   "email": "sam@example.com",
40   "password": "Password@12345"
41 }
42
43 Send Request
44
45 GET http://localhost:5095/api/admin/dashboard
46 Content-Type: application/json
47
48 {
49   "fullName": "Frank Sam",
50   "email": "sam@example.com",
51   "password": "Password@12345"
52 }
```

```
1 HTTP/1.1 200 OK
2 Connection: close
3 Content-Type: application/json; charset=utf-8
4 Date: Sat, 30 May 2026 17:25:43 GMT
5 Server: Kestrel
6 Transfer-Encoding: chunked
7
8 {
9   "email": "sam@example.com",
10  "roles": []
11 }
```



```
10 GET http://localhost:5095/api/admin/dashboard
11 Content-Type: application/json
12 Authorization: Bearer
13
14 Send Request
15
16 POST http://localhost:5095/api/auth/login
17 Content-Type: application/json
18
19 {
20   "fullName": "Frank Sam",
21   "email": "sam@example.com",
22   "password": "Password@12345"
23 }
24
25 Send Request
26
27 GET http://localhost:5095/api/user/profile
28 Content-Type: application/json
29
30 {
31   "fullName": "Frank Sam",
32   "email": "sam@example.com",
33   "password": "Password@12345"
34 }
35
36 Send Request
37
38 GET http://localhost:5095/api/admin/dashboard
39 Content-Type: application/json
40
41 {
42   "fullName": "Frank Sam",
43   "email": "sam@example.com",
44   "password": "Password@12345"
45 }
```

```
1 HTTP/1.1 200 OK
2 Connection: close
3 Content-Type: application/json; charset=utf-8
4 Date: Sat, 30 May 2026 17:19:16 GMT
5 Server: Kestrel
6 Transfer-Encoding: chunked
7
8 {
9   "email": "sam@example.com",
10  "roles": [
11    "Admin"
12  ]
13 }
```



```
10 GET http://localhost:5095/api/admin/dashboard
11 Content-Type: application/json
12 Authorization: Bearer
13
14 Send Request
15
16 POST http://localhost:5095/api/auth/login
17 Content-Type: application/json
18
19 {
20   "fullName": "Frank Sam",
21   "email": "sam@example.com",
22   "password": "Password@12345"
23 }
24
25 Send Request
26
27 GET http://localhost:5095/api/user/profile
28 Content-Type: application/json
29
30 {
31   "fullName": "Frank Sam",
32   "email": "sam@example.com",
33   "password": "Password@12345"
34 }
35
36 Send Request
37
38 GET http://localhost:5095/api/admin/dashboard
39 Content-Type: application/json
40
41 {
42   "fullName": "Frank Sam",
43   "email": "sam@example.com",
44   "password": "Password@12345"
45 }
```

```
1 HTTP/1.1 200 OK
2 Connection: close
3 Content-Type: application/json; charset=utf-8
4 Date: Sat, 30 May 2026 17:39:16 GMT
5 Server: Kestrel
6 Transfer-Encoding: chunked
7
8 {
9   "email": "sam@example.com",
10  "roles": [
11    "Admin"
12  ]
13 }
```